Linear Algebra

Assignment - 2

2) $$A = \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 2 & 13 & 7 & -15 & 21 \\ -1 & 7 & 19 & -24 & 12\cancel{8} \\ -3 & -15 & -24 & 54 & 18 \\ -3 & 21 & 12 & 18 & 126 \end{bmatrix}$$

① LDL^T Decomposition

$$A = LDL^T$$

- $L \Rightarrow$ Lower matrix
- $D \Rightarrow$ Diagonal matrix
- $L^T \Rightarrow$ transpose of L

Let

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ L_{21} & 1 & 0 & 0 & 0 \\ L_{31} & L_{32} & 1 & 0 & 0 \\ L_{41} & L_{42} & L_{43} & 1 & 0 \\ L_{51} & L_{52} & L_{53} & L_{54} & 1 \end{bmatrix}$$

$\therefore d_1 = 1$

$\therefore L_{21} = \dfrac{a_{21}}{d_1} = \dfrac{2}{1} = 2$

$\therefore L_{31} = \dfrac{a_{31}}{d_1} = \dfrac{-1}{1} = -1$

$\therefore L_{41} = \dfrac{a_{41}}{d_1} = \dfrac{-3}{1} = -3$

$\therefore L_{51} = \dfrac{a_{51}}{d_1} = \dfrac{-3}{1} = -3$

Now, $L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -3 & 0 & 0 & 1 & 0 \\ -3 & 0 & 0 & 0 & 1 \end{bmatrix}$

$\therefore A' = A - L_1 D_1 L_1^T$

$= A - \begin{bmatrix} 1 \\ 2 \\ -1 \\ -3 \\ -3 \end{bmatrix} \cdot 1 \cdot \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \end{bmatrix}$

$= A - \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 2 & 4 & -2 & -6 & -6 \\ -1 & -2 & 1 & 3 & 3 \\ -3 & -6 & 3 & 9 & 9 \\ -3 & -6 & 3 & 9 & 9 \end{bmatrix}$

$\therefore$ After element wise subtraction

$A' = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 9 & -9 & 27 \\ 0 & 9 & 18 & -27 & 9 \\ 0 & -9 & -27 & 45 & 9 \\ 0 & 27 & 9 & 9 & 117 \end{bmatrix}$

$\therefore d_2 = 9$

Now,

$L_{32} = \dfrac{A'_{32}}{d_2} = \dfrac{9}{9} = 1$

$L_{42} = \dfrac{A'_{42}}{d_2} = \dfrac{-9}{9} = -1$

$L_{52} = \dfrac{A'_{52}}{d_2} = \dfrac{27}{9} = 3$

Now, $L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ -3 & -1 & 0 & 1 & 0 \\ -3 & 3 & 0 & 0 & 1 \end{bmatrix}$

$\therefore$ Updating A' again,

$$A'' = A' - L_2 D_2 L_2^T$$

$$= A' - \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \\ 3 \end{bmatrix} \cdot 9 \cdot \begin{bmatrix} 0 & 1 & 1 & -1 & 3 \end{bmatrix}$$

$$= A' - \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 9 & -9 & 27 \\ 0 & 9 & 9 & -9 & 27 \\ 0 & -9 & -9 & 9 & -27 \\ 0 & 27 & 27 & -27 & 81 \end{bmatrix}$$

$$A'' = \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 2 & 0 & 0 & 0 & 0 \\ -1 & 0 & 9 & -18 & -18 \\ -3 & 0 & -18 & 36 & 36 \\ -3 & 0 & -18 & 36 & 36 \end{bmatrix}$$

$\therefore$ $d_3 = 9$

$\therefore$ $L_{43} = \dfrac{A''_{43}}{d_3} = \dfrac{-18}{9} = -2$

$L_{53} = \dfrac{A''_{53}}{d_3} = \dfrac{-18}{9} = -2$

Now, $L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ -3 & -1 & -2 & 1 & 0 \\ -3 & 3 & -2 & 0 & 1 \end{bmatrix}$

∴ Updating A again

$$A'''^{T} = A'' - L_3 D_3 L_3^T$$

$$= A'' - \begin{bmatrix} 0 \\ 0 \\ 1 \\ -2 \\ -2 \end{bmatrix} \cdot 9 \cdot \begin{bmatrix} 0 & 0 & 1 & -2 & -2 \end{bmatrix}$$

$$= A'' - \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 9 & -18 & -18 \\ 0 & 0 & -18 & 36 & 36 \\ 0 & 0 & -18 & 36 & 36 \end{bmatrix}$$

$$A''' = \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 2 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ -3 & 0 & 0 & 0 & 0 \\ -3 & 0 & 0 & 0 & 0 \end{bmatrix}$$

∴ $L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ -3 & -1 & -2 & 1 & 0 \\ -3 & 3 & -2 & 0 & 1 \end{bmatrix}$ & $D = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore$ Let's verify $A = L \cdot D \cdot L^T$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ -3 & -1 & -2 & 1 & 0 \\ -3 & 3 & -2 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 0 & 1 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & -1 & -3 & -3 \\ 2 & 13 & 7 & -15 & 21 \\ -1 & 7 & 19 & -24 & 12 \\ -3 & -15 & -24 & 54 & 18 \\ -3 & 21 & 12 & 18 & 126 \end{bmatrix}$$

(B) By cholesky decomposition

$L_{11} = \sqrt{A_{11}} = 1$

$L_{21} = \dfrac{A_{21}}{L_{11}} = \dfrac{2}{1} = 2$

$L_{22} = \sqrt{A_{22} - L_{21}^2} = \sqrt{13 - 2^2} = \sqrt{9} = 3$

$L_{31} = \dfrac{A_{31}}{L_{11}} = \dfrac{-1}{1} = -1$

$L_{32} = \dfrac{A_{32} - l_{31} L_{21}}{L_{22}} = \dfrac{7 - (-1 \times 2)}{3} = \dfrac{9}{3} = 3$

$L_{33} = \sqrt{A_{33} - L_{31}^2 - L_{32}^2} = \sqrt{19 - 1 - 9} = \sqrt{9} = 3$

$L_{41} = \dfrac{A_{41}}{L_{11}} = \dfrac{-3}{1} = -3$

$L_{42} = \dfrac{A_{42} - L_{41} L_{21}}{L_{22}} = \dfrac{-15 - (-3 \times 2)}{3} = \dfrac{-9}{3} = -3$

$$L_{43} = \frac{A_{43} - L_{41}L_{31} - L_{42}L_{32}}{L_{33}} = \frac{-24-3+9}{3} = \frac{-18}{3} = -6$$

$$L_{44} = \sqrt{A_{44} - L_{41}^2 - L_{42}^2 - L_{43}^2} = \sqrt{54-9-9-36} = \sqrt{0} = 0$$

$$L_{51} = \frac{A_{51}}{L_{11}} = \frac{-3}{1} = -3$$

$$L_{52} = \frac{A_{52} - L_{51}L_{21}}{L_{22}} = \frac{21+6}{3} = \frac{27}{3} = 9$$

$$L_{53} = \frac{A_{53} - L_{51}L_{31} - L_{52}L_{32}}{L_{33}} = \frac{12-3-27}{3} = \frac{-18}{3} = -6$$

$$L_{54} = \frac{A_{54} - L_{51}L_{41} - L_{52}L_{42} - L_{53}L_{43}}{L_{33}} = 0 \quad (\because L_{44}=0)$$

$$\therefore L = \begin{bmatrix} \cancel{1} & \cancel{0} & \cancel{0} & 0 & 0 \\ \cancel{L_{41}} & \cancel{1} & \cancel{0} & \cancel{0} & 0 \\ \cancel{L_{21}} & \cancel{L_2} & & & \end{bmatrix}$$

$$\therefore L = \begin{bmatrix} L_{11} & 0 & 0 & 0 & 0 \\ L_{21} & L_{22} & 0 & 0 & 0 \\ L_{31} & L_{32} & L_{33} & 0 & 0 \\ L_{41} & L_{42} & L_{43} & L_{44} & 0 \\ L_{51} & L_{52} & L_{53} & L_{54} & L_{55} \end{bmatrix}$$

$\therefore$ Now, putting the values

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 \\ -1 & 3 & 3 & 0 & 0 \\ -3 & -3 & -6 & 0 & 0 \\ -3 & 9 & -6 & 0 & 0 \end{bmatrix}$$

As we require Diagonal matrix as 1's